



Military Institute of Science and Technology

Department of Computer Science and Engineering

CSE-464 (Software Development Project II)

CampusNet: An Interactive University Community Platform

Final Project Report

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I. INTRODUCTION

CampusNet is an interactive mobile application built using Flutter. The app connects students, alumni and faculty members, creating a unified platform for campus life. It provides essential services such as an AI-powered chatbot, campus navigation map, messaging system and emergency features such as blood donation ambulance services etc. The app aims to help new students connect with their peers and alumni and help them navigate the campus more effectively.

This application addresses several pain points for students, particularly for freshers, such as finding alumni expertise, navigating the campus and accessing critical services. Through features like profile generation, tuition service opportunities and emergency features, CampusNet aims to provide a seamless experience for both students and alumni.

II. TARGET USERS

The app is designed for a wide range of users including:

- Undergraduate Students
- Graduate Students
- Alumni

III. LITERATURE REVIEW OF EXISTING CAMPUS APPS

Several universities have developed mobile applications to enhance student engagement, streamline administrative processes and provide essential campus services. Below is an overview of five such applications:

1) Campus.pk

A comprehensive e-learning platform offering video lectures, MCQ tests, online notes and live sessions for students across Pakistan. It aims to provide flexible learning opportunities and facilitate teacher-student interactions. *[Available on Google Play Store]*.

2) Campus+ by FH Campus Wien

Designed for students at FH Campus Wien, this app provides access to class schedules, campus card balances, cafeteria menus, course overviews and news updates. It serves as a central hub for academic and campus-related information. *[Available on Google Play Store]*.

3) Campus Student

A widely used app in the U.S., Campus Student offers real-time access to announcements, assignments, attendance, grades and schedules. It aims to keep students informed and organized throughout their academic journey. *[Available on Apple App Store]*.

4) Campus.uno

This platform provides a comprehensive ERP solution for educational institutions, facilitating access to academic information for students, parents and educators. It aims to streamline communication and administrative processes within educational institutions. *[Available on Google Play Store]*.

5) Campus+ (UM6P)

Developed for Mohammed VI Polytechnic University, this app offers access to campus events, news, class schedules and program information. It aims to enhance the university experience by providing students with relevant information at their fingertips. *[Available on Apple App Store]*.

TABLE I
COMPARISON OF EXISTING CAMPUS APPS

App	Key Features	Limitations
Campus.pk	Video lectures, MCQ tests, live sessions	Limited to Pakistan; lacks campus-specific services
Campus+ (FH Campus Wien)	Class schedules, campus card balance, cafeteria menu	Limited to FH Campus Wien students
Campus Student	Real-time access to grades, assignments, attendance	Requires institutional login; U.S.-centric
Campus.uno	ERP solution for academic and administrative processes	Primarily an administrative tool; lacks student engagement features
Campus+ (UM6P)	Campus events, news, class schedules	Limited to UM6P students

IV. PROBLEM STATEMENT AND IDENTIFICATION

Based on the review of existing systems, the primary issues identified are:

- Lack of a unified platform for campus-related services.
- Inadequate connection mechanisms for students and alumni.
- Fragmentation of essential services like campus navigation, emergency services and tuition opportunities across different apps.
- Lack of AI-driven support for campus-related questions.

V. OBJECTIVES

The main objectives of the CampusNet app are:

- To provide instant assistance and campus guidance through an integrated AI chatbot.
- To build a unified platform for communication, emergency services and alumni-student networking.
- To enhance campus navigation and accessibility with an interactive map.

VI. FEATURE LIST

The core features of the app include:

- **Chatbot (NetBot):** AI-powered assistant for answering questions and giving campus info.
- **Campus Map:** Interactive map to help users navigate campus buildings, classrooms and auditoriums.
- **Blog Post Feed:** Platform for students and clubs to share updates, events and stories.
- **Resource Bank:** A centralized repository of study materials, including notes, textbooks and other academic resources for all courses across departments.
- **Tuition Services:** A platform to connect tutors and learners.
- **Messaging:** One-to-one and group messaging with resource sharing.
- **Emergency Services:** Features to request and donate blood and contact emergency services like ambulances.
- **Profile and CV Generator:** A feature to create and share a student's profile and generate a CV.

VII. TECHNOLOGY STACK

The following technologies were used in the development of CampusNet:

TABLE II
TECHNOLOGY STACK

Component	Tools Used
Frontend Framework	Dart
Backend Framework	Flask
Database	PostgreSQL
Additional Tools	Grok API

VIII. DEVELOPED SYSTEM

Below are some key features of the app:

- **Landing Page:** The first screen users see, showing available services and features. Figure 9 shows the landing page.
- **Profile Page:** Displays the user's personal and academic information.
- **NetBot Chatbot:** A page where users can interact with the AI chatbot. Figure 2 illustrates the emergency section.
- **Campus Map:** A page that helps users navigate the campus. Figure 10 illustrates the Campus Map feature.
- **Emergency Services:** A page for requesting and donating blood or contacting ambulance services. Users can call ambulance, find blood donor, request for blood, see all blood requests and can register as blood donor. Figure 7 illustrates the emergency section.

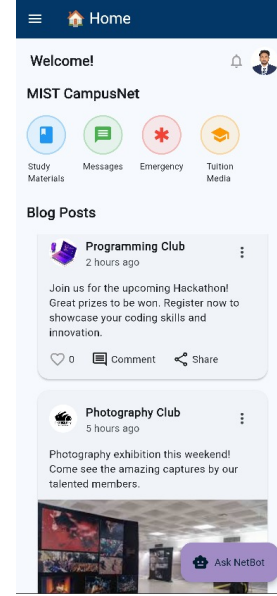


Fig. 1. Landing Page

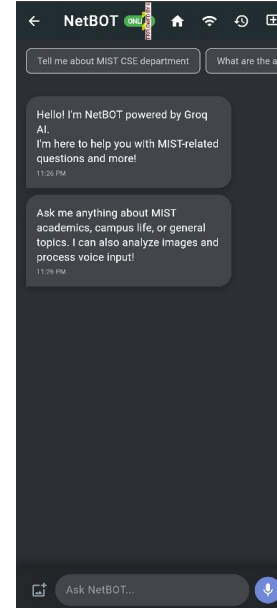


Fig. 2. Chatbot features.

- **Tuition:** Here user can look for tutors based on requirements and also offer for tuition. Figure 4 illustrates the tuition section.
- **Message:** Here user can send messages to other, can create group, share documents. Figure 5 illustrates the message feature.
- **Resource Bank:** A centralized library for academic materials. Users can browse or upload resources by department and course. Within each course, chapters organize content, users can add new courses (in

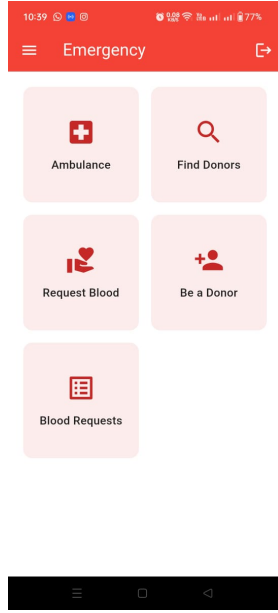


Fig. 3. Emergency features.

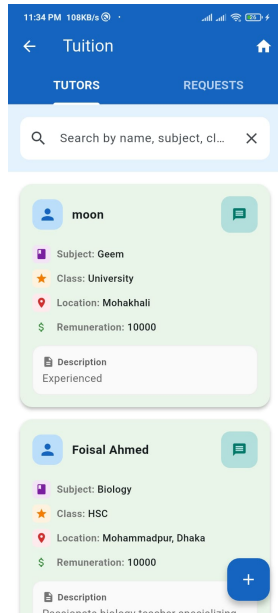


Fig. 4. Tuition features.



Fig. 5. Message features.

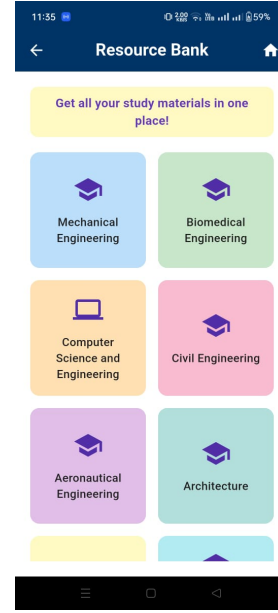


Fig. 6. Resource Bank feature.

their department), upload PDFs and other files and read materials in-app. Deletion of a file is restricted to its original uploader / owner. Figure 6 illustrates the message feature.

IX. ETHICAL CONCERNS

The app raises several ethical concerns, including:

- User data (personal, academic) is collected and stored securely with user consent.

- Location data is used exclusively for campus navigation with explicit user approval.
- Blood donation and emergency service features require user consent to share contact details.
- No microphone or camera access is requested unless the user voluntarily uses the voice input feature for the chatbot.

X. SOCIETAL IMPACTS

CampusNet can have a significant positive impact:

- Helps freshers navigate campus and connect with peers quickly.
- Fosters a sense of community by enabling alumni-student mentorship.
- Promotes social responsibility through features like blood donation and tutoring opportunities.
- Enables students to access and share study materials/resources effortlessly.

XI. DESIGN CHANGES

After faculty feedback, the following changes were implemented:

- Organized emergency features into a single section for better accessibility. Figure 7 illustrates the re-designed emergency section.

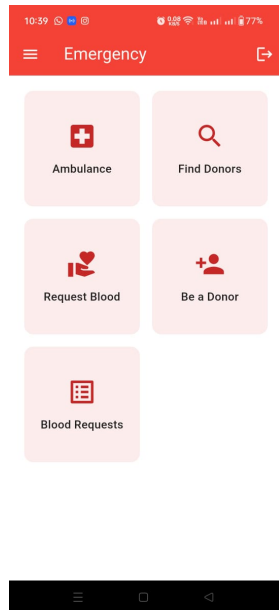


Fig. 7. Emergency features consolidated into one section.

- Improved the dashboard design for better usability and visual clarity. Improved visual hierarchy in blog posts for better readability. Figure 8 shows the previous version, while Figure 9 presents the updated dashboard.
- Implemented map integration. Figure 10 illustrates the Campus Map feature.
- Improved UI for the Resource Bank by adding a search option and enhancing the color scheme. Figure 11 shows the initial version, while Figure 12 presents the improved version.
- Added an ambulance-request feature; Figure 13 shows the ambulance page.

XII. LIFELONG LEARNING

Using the app, students and alumni can:

- Continuously interact even after graduation, ensuring lasting connections.
- Apply app development knowledge in future real-life projects.
- Engage in lifelong learning through mentoring and knowledge-sharing opportunities.

XIII. SUS USABILITY TESTING SCORE AND AVERAGE TASK COMPLETION TIME

The System Usability Scale (SUS) evaluation yielded a score of 92.5, which falls in the “A / Excellent”

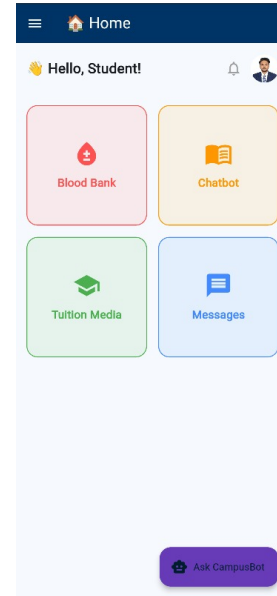


Fig. 8. Previous Dashboard

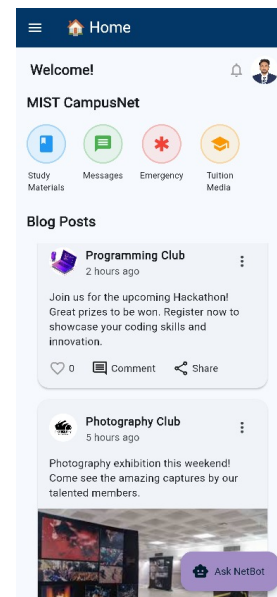


Fig. 9. Current Dashboard

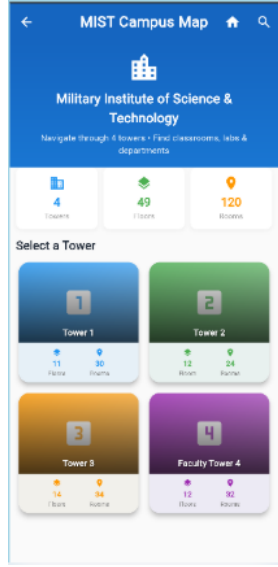


Fig. 10. Campus Map interface.

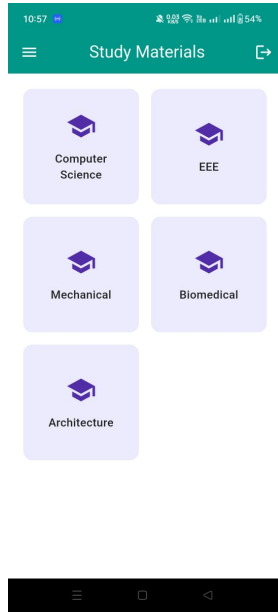


Fig. 11. Initial Resource Bank interface.

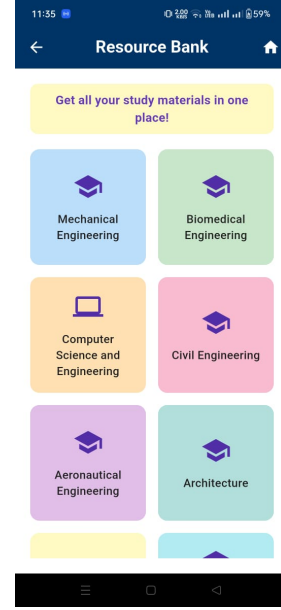


Fig. 12. Improved Resource Bank interface with search option and color enhancements.

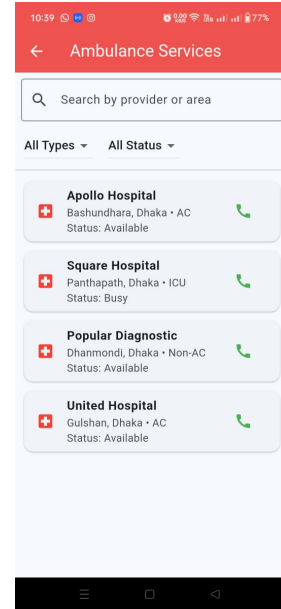


Fig. 13. Ambulance-request page.

range. The three benchmark tasks were: Map navigation, Emergency request and Tuition/resources. The average task completion time was 26 s based on two evaluators. The Emergency task was completed the fastest, Map was intermediate and Tuition/resources took the longest, suggesting minor navigation overhead in the resources flow. Detailed per-task timings for each evaluator are summarized in Table III.

XIV. FUTURE WORK

The following features are planned for future updates:

TABLE III
TASK COMPLETION TIME (SECONDS) BY EVALUATOR AND AVERAGE.

Task Name	Evaluator 1	Evaluator 2	Average
Map (Campus navigation)	26.48	22.21	23.95
Emergency (request flow)	15.70	18.27	16.985
Tuition / Resources	31.23	36.37	33.50
Overall Average	26.69		

- AI-driven skill matching between students and alumni.
- Analytics for monitoring engagement and activity on the platform.
- Implementing dark mode and accessibility improvements.
- UI improvements for better user experience.
- Deployment of the platform for real-world use.

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